

End User Operations Manual

Step-by-Step Operational Guide to Account Management, AI Price Predictions & Dashboard Analytics

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Authoring Team:	Narendar Reddy, Manvitha, Pravallika, Ashwindh
Supported Browsers:	Google Chrome 120+, Microsoft Edge 120+, Mozilla Firefox 120+

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1.0 DOCUMENT REVISION HISTORY

Rev #	Date	Author	Reviewer	Approver	Description of Changes
1.0	2026-08-01	Pravallika	Narendar R.	Infosys Mentor	Initial End User Operating Manual Draft
1.5	2026-08-04	Pravallika	Manvitha	Technical Lead	Added OTP Password Recovery & Prediction Instructions
2.0	2026-08-07	Team PricePilot	QA Team	Academic Committee	Final User Operations Manual Validation

2.0 INTRODUCTION & PLATFORM OVERVIEW

Welcome to **PricePilot AI**. PricePilot AI is an artificial intelligence-powered dynamic pricing and demand forecasting platform designed to help e-commerce merchants optimize selling prices in real time. Powered by an Extra Trees machine learning model (96.50% precision), PricePilot AI computes optimal product prices, estimated costs, profit margin percentages, and confidence scores based on shipping freight value, physical weight, dimensions, and category parameters.

3.0 SYSTEM ACCESS & WEB BROWSERS

To access PricePilot AI, open a supported HTML5 web browser and navigate to the application URL:

- **Production Access URL:** ``https://pricepilot-ai.vercel.app``
- **Local Development URL:** ``http://localhost:5173``
- **Supported Browsers:** Google Chrome 120+, Microsoft Edge 120+, Mozilla Firefox 120+, Apple Safari 17+.

4.0 ACCOUNT REGISTRATION & ONBOARDING

To create a new PricePilot AI account:

1. Open ``/login`` page and click the **'Create Account'** tab.
2. Enter your Full Name, Email Address, Unique Username, Phone Number, and Password.
3. Click **'Sign Up'**. Upon submission, your account will be registered with *'pending'* approval status until approved by a System Administrator.

[SCREENSHOT PLACEHOLDER: Account Registration Form UI]

5.0 SECURE ACCOUNT LOGIN

To log in to your account:

1. Navigate to `/login``.
2. Enter your registered Username or Email and Password.
3. Click **'Sign In'**. Upon successful verification, an OAuth2 Bearer JWT access token is stored securely, and you are automatically redirected to the `/dashboard``.

[SCREENSHOT PLACEHOLDER: Secure Login Form UI]

6.0 6-DIGIT OTP PASSWORD RESET & ACCOUNT RECOVERY

If you forget your password:

1. On the Login screen, click **'Forgot Password?'**.
2. Enter your registered Email or Phone Number and click **'Send OTP'**.
3. A **6-digit numeric OTP code** (valid for 15 minutes) is generated.
4. Enter the 6-digit OTP code along with your new password and click **'Reset Password'**.

[SCREENSHOT PLACEHOLDER: 6-Digit OTP Password Reset Modal UI]

7.0 INTERACTIVE DASHBOARD OVERVIEW

The Dashboard (`/dashboard``) serves as the central command hub displaying key operational metrics:

- **Total Predictions Card:** Shows total AI price calculations executed.
- **Average Profit Margin Card:** Displays real-time estimated profit margin percentage.
- **System Health Status Card:** Displays database and ML model connection health.
- **Quick Action Buttons:** Single-click shortcuts to New Prediction, Analytics, and Docs.

[SCREENSHOT PLACEHOLDER: Main Dashboard View UI]

8.0 AI PRICE PREDICTION CALCULATOR GUIDE

To execute an AI price prediction:

1. Navigate to **'New Prediction'** (`/predict`).
2. Enter the Product Category (1–73 taxonomy), Shipping Freight Value (■), Product Weight (g), Length (cm), Height (cm), and Width (cm).
3. Click **'Calculate Price'**.
4. The Extra Trees ML model runs inference (<45ms) and displays:

Output Parameter	Sample Value	Operational Meaning & Interpretation
Predicted Price	■ 189.50	Optimal AI-recommended selling price to maximize revenue.
Estimated Cost	■ 123.18	Estimated item cost structure derived from product attributes.
Profit Margin %	35.0 %	Estimated gross profit margin percentage.
Confidence Score	96.5 %	ML model confidence score based on Extra Trees R ² benchmark.
Demand Level	High Demand	Forecasted market demand volume tier.

[SCREENSHOT PLACEHOLDER: AI Prediction Calculator & Result Card UI]

9.0 PREDICTION HISTORY & SYSTEM ANALYTICS

- **Prediction History** (`/history`): Review historical price calculations, filter by date or category, and inspect input feature details.
- **Analytics** (`/analytics`): View visual demand distribution charts, category revenue benchmarks, and profit margin trends.

10.0 PROFILE MANAGEMENT & THEME CUSTOMIZATION

- **Profile Page** (`/profile`): View your account role, update Full Name, Phone Number, and Avatar URL.
- **Settings Page** (`/settings`): Toggle Dark Glassmorphism theme, adjust notification preferences, and select interface language.

11.0 FREQUENTLY ASKED QUESTIONS (FAQ)

#	Frequently Asked Question	Detailed Answer
1	How accurate are the price predictions?	Predictions are powered by an Extra Trees Regressor achieving 96.50% R ² accuracy.
2	Why is my account status 'pending'?	New user registrations require Admin approval before full system access is granted.
3	How long does a password reset OTP remain valid?	6-digit OTP codes remain active for 15 minutes from generation.
4	Can I export my user registry to Excel?	Yes, Administrators can export native .xlsx reports via the `/users` page.

12.0 TROUBLESHOOTING & ERROR RESOLUTION

Error Condition	Probable Cause	Recommended Resolution Action
'Invalid Credentials' (401)	Incorrect email or password	Re-enter password or use 'Forgot Password' OTP reset.
'Account Pending Approval' (403)	User account not approved by Admin	Contact system administrator to approve account status.
'Model Not Loaded' (503)	ML model binary file missing	Ensure `trained_models/best_price_model.pkl` exists on server.
'Network Connection Error'	Backend server offline	Verify backend API running at `http://localhost:8000`.

13.0 OPERATIONAL BEST PRACTICES

1. Always input accurate shipping weight (g) and volume (cm³) parameters to ensure maximum price prediction precision.
2. Regularly review Prediction History logs to monitor category pricing trends.
3. Log out securely after completing administrative or prediction operations.